



Transforming E-commerce: A Microservices-Based Taxonomy for Cloud Applications

Waleed Ibrahim^{1,2,3(✉)} and Van Kiet Luong⁴

¹ The University of Sydney, Business School, Darlington, Australia
waleed.ibrahim@sydney.edu.au

² Central Queensland University, Rockhampton, Australia

³ Victoria University, Melbourne, Australia

⁴ Western Sydney University, Sydney, Australia

Abstract. The rapid evolution of e-commerce is significantly influenced by the integration of microservices and cloud computing, yet a complex set of challenges arises from this integration, underscoring a critical gap in streamlined implementation and management. Existing studies have provided insights into individual aspects of this integration but lack a comprehensive approach to effectively navigate the intricacies involved. There is a need for a holistic framework that encompasses not only the technological aspects but also the strategic, operational, and managerial facets essential for optimizing e-commerce platforms. This research intends to fill this gap by offering a detailed taxonomy tailored for microservices architecture in cloud-based e-commerce platforms, ensuring businesses can seamlessly integrate, manage, and optimize these technologies to enhance scalability, flexibility, and customer experience amidst the rapidly changing digital commerce landscape.

Keywords: E-Commerce Evolution · Microservices Architecture · Cloud Based Applications · Monolithic architecture · Microservices Integration

1 Introduction

E-commerce has undergone a transformation over the past decade, thanks to technological advancements that have made it a key player in the world economy. Technology advancements have not only streamlined business operations but have also improved the entire consumer experience. Enhanced connectivity to the internet, the development of secure online payment ecosystems, and evolving consumer demands for convenient, personalised shopping experiences have all played an important part in shifting this industry forward.

The widespread adoption of cloud-based applications and services has been one of the most significant technical developments in this field, making the process more flexible and scalable, as noted by [1]. Unlike before, retailers no longer need to make

heavy upfront investments in website hosting, inventory management, and customer data analytics, as cloud technologies offer the required resources on demand.

The advent and emergence of the microservices architecture have accelerated the evolution of e-commerce even more. According to [2], this architecture has increased the flexibility and adaptability of online retail platforms by developing applications as a collection of loosely coupled, independently deployable services. The shift from the long-standing traditional monolithic architecture has been proven to be important, as it struggled to adapt to the rapidly evolving and dynamic e-commerce industries. In addition, Velepucha and Flores [3] discussed the inherent issues of scalability, maintainability, and agility associated with monolithic designs. The microservices architecture, with its modular nature, effectively addresses these challenges. E-commerce businesses can benefit from enhanced scalability, simplified maintenance processes, and increased agility for maintaining competitiveness and responsiveness in a marketplace defined by constant evolution.

The synthesis of cloud computing and e-commerce has certainly provided businesses with the ability to dynamically access resources, scale operations with ease, and ensure consistent customer experiences across different places. While microservices architecture provides a compelling solution to e-commerce system's scalability and agility concerns, it also brings substantial implementation challenges. These include managing distributed systems, ensuring data consistency, and maintaining security in a cloud-based environment. Given these challenges, the study provides a taxonomy that categorises microservices into separate functional groups, aimed at addressing these challenges in a systematic way.

The objective of this research is to develop a comprehensive taxonomy for microservices architecture in cloud-based e-commerce applications. The framework will help businesses in organising and navigating the complexities of microservices. This also help them to fully realise the advantages while managing possible issues. This goal is particularly pivotal in an era where e-commerce is not just about selling products online but has morphed into a complex, multifaceted ecosystem.

2 Microservices and Cloud Computing

This section provides a comprehensive overview of key concepts and research findings related to microservices architecture, cloud computing in e-commerce, and adaptive microservices for dynamic e-commerce systems. It aims to connect these findings and highlight their importance in the context of modernising e-commerce system.

2.1 Microservices Architecture

Microservices architecture represents an important transformation in the design, development, and deployment of software programmes. It is characterized by breaking down monolithic applications into smaller, self-contained services that can be built, deployed, and scaled independently. This design architectural has gained significant attention in recent year because of its potential to address the limitations of monolithic architectures. Velepucha and Flores [3] conducted a comprehensive survey exploring the principles,

patterns, and migration challenges associated with microservices architecture. They emphasize the importance of principles such as domain-driven design, replaceability, and autonomous development teams in guiding the successful migration from monolithic systems to microservice. This study contributes to the understanding of microservices' foundational principles and their connection to both monolithic and microservices architectures.

The study [4] provides additional insights into the evolution of application architectures, focusing on the shift from traditional monolithic structures to Service-Oriented Architecture (SOA), and microservices. They emphasise the importance of microservices in breaking down monolithic applications into fine-grained, independently designed, tested, and deployed services. Container technology, as highlighted by [5], is critical in effectively managing microservices by encapsulating and isolating these services for simpler administration and scaling.

2.2 Monolithic and Microservices Architectures Comparison

In the e-commerce landscape, choosing between microservices and traditional monolithic architectures requires careful consideration of various factors. Monolithic architectures, characterized by their single, tightly coupled codebase. However, when e-commerce systems increase in size, they frequently face scalability and maintenance issues. Microservices provide an alternative approach by decomposing monolithic applications into smaller, self-contained services [6]. This decomposition results in fine-grained services, each focusing on a specific business capability. Scalability is one of the most major advantages of microservices. E-commerce platforms often experience fluctuating traffic patterns, especially during promotions or seasonal events. Microservices enable organisations to scale individual services, ensuring that resources are deployed efficiently to manage workload properly.

However, microservices introduce complexities related to inter-service communication, data consistency, and debugging in distributed environments [6]. These challenges must be carefully addressed. Furthermore, the shift from a monolithic to a microservices architecture, as supported in the proposed framework by [6], Monolith to Microservices Decomposition Framework (M2MDF), requires a well-structured design and thorough validation in real-world settings. Therefore, the choice between monolithic and microservices architecture in the context of e-commerce should be aligned with organisations' growth path, scalability requirements, and development environment. While microservices offer agility and scalability, monolithic architecture might be suitable for smaller e-commerce businesses with limited resources.

2.3 Cloud Computing in E-commerce

Cloud computing integration in the e-commerce sector is a disruptive trend changing the digital economy. Almarabeh and Majdalawi [1] highlight the efficiency and cost-effectiveness of cloud computing, which has offered enterprises a wide range of solutions for business operational tasks. They emphasize the potential benefits and challenges of cloud computing in e-commerce, such as scalability, cost-effectiveness, and resource

flexibility. This aligns with the idea that cloud technology can optimize resources, particularly for small and medium-sized enterprises, allowing them to compete effectively with larger firms.

Moreover, Kumari and Kumar. S [7] present a cloud-native architecture for real-time pricing in e-commerce. Real-time pricing, a dynamic pricing technique, uses the capabilities of cloud computing to update prices in response to changes in demand and competition. The authors stress the significance of cloud-native technology in handling real-time pricing challenges such as real-time decision-making and managing massive data volumes. This highlights the role of cloud infrastructure in optimizing pricing strategies, as well as highlighting the integration of cloud computing and e-commerce.

2.4 Adaptive Microservices for Dynamic E-commerce

Adaptive microservices for dynamic e-commerce platforms are the next step in modernising e-commerce systems. Dobrița [8] investigates the use of adaptive microservices architecture to provide personalized experiences using machine learning and real-time adaption. This aligns with the evolving role of microservices in e-commerce, where self-adaptive systems and personalized product recommendations are more important.

The research [8] emphasizes the importance of machine learning in e-commerce platforms, with applications ranging from recommendation systems to chatbots, fraud detection, and dynamic pricing. Machine learning and artificial intelligence are essential components of modern e-commerce operations, addressing the demand for personalised shopping experience. In addition, the research proposes self-adaptive systems and the MAPE-K loop model for continuous monitoring and adaptation. The MAPE-K (Monitor, Analyse, Plan, Execute, and Knowledge) is an autonomic management paradigm that is essential for adaptable microservices. This concept enables systems to self-manage by continually monitoring their performance, analysing data, planning and executing real-time modifications, and using learned insights to optimise future responses [8]. This highlights how adaptive microservices, machine learning, and real-time adaption work together to improve the dynamic nature of e-commerce platforms.

2.5 Challenges and Limitations of Microservices in E-commerce

Despite the advantages of microservices, there are various obstacles and limitations to consider when using them in cloud-based e-commerce systems. A systematic review was conducted by [6], who highlighted numerous important issues and limitation. Firstly, breaking down monolithic applications into microservices can be a time-consuming and resource-intensive operation. Organisations may face challenges in identifying appropriate boundaries for microservices, ensuring proper data consistency, and managing the migration of existing systems. Secondly, because microservices have a distributed architecture, they might increase the complexity of monitoring and debugging. To solve these problems, Abgaz et al. [6] have emphasized the importance of good monitoring and logging system. Moreover, ensuring the security of microservices and their interactions is crucial in e-commerce, where sensitive customer data is often processed. Abgaz et al. [6] have also stated the importance of strong security and identity management inside a microservices ecosystem. Scalability is another challenge, as e-commerce platforms

often experience fluctuating traffic loads. According to the research [4], microservices should be designed to scale horizontally to suit increased demand. Furthermore, maintaining data consistency and transactional integrity across several microservices can be difficult. To solve these problems, the findings of [1] suggest investigating event-driven systems and distributed transaction patterns.

In response to the complications of microservices deployment, particularly in cloud-based e-commerce, this study proposes a taxonomy to address these challenges. Microservices, while promising in terms of flexibility and scalability, introduce issues such as service integration, data consistency, and transaction management. Our taxonomy provides a structured approach to these challenges by clearly defining service boundaries, suggesting data management patterns, and proposing robust communication protocols between services, allowing for more efficient implementation and management of microservices architectures.

2.6 Previous Works and Taxonomies Related to Microservices

Previous research, such as that of [3] and [9], has significantly contributed to understanding and deploying microservices, particularly in the context of e-commerce. Their research explores the principles and patterns of microservices architecture, its applications in scalable systems, and challenges of transitioning from monolithic to microservices architecture. The systematic review and the development of the M2MDF give significant insights and direction for organisations attempting to move from monolithic to microservices architectures [6]. This methodology provides an organized way to solve the issues of monolith breakdown, allowing for a more seamless adoption of microservices. However, the M2MDF framework did not adequately address the challenge of scaling microservices independently during high-demand periods in e-commerce, such as during flash sales or promotional events like Black Friday. In addition, the research [5] provides frameworks for containerized microservices orchestration, which align with the broader usage of containerization technologies [5]. These frameworks offer ways for managing microservice execution efficiently, optimising resource utilisation, and reacting to changing workloads [5]. Yu et al. [10] pointed out a lack of standards for orchestrating microservices, but they failed to address the security risks associated with managing numerous containers in a distributed environment, particularly when a cloud-based e-commerce system handles sensitive customer data. Furthermore, the paper [11] offers an extensive taxonomy of containerization technologies and explores their applications across various domains, including scientific computing. Their work underscores the role of containerization in enabling the flexible deployment of high-performance systems, aligning with the project's focus on optimizing e-commerce systems through microservices and containerization. These works help to provide comprehensive techniques and methodologies for deploying and managing microservices in e-commerce applications. They serve as valuable references for businesses aiming to leverage microservices to enhance their e-commerce platforms and customer experiences.

3 Methodology

3.1 Objective

The primary objective of this study is to clarify the scope and impact of microservices architecture in cloud-based e-commerce through an analysis of existing literature. The review will provide insights into current trends, challenges, and future prospects, serving as a fundamental resource for further empirical research.

3.2 Identification of Research

A comprehensive and exhaustive search approach has been developed to search through a wide range of existing literature and ensure the finding of essential data. The strategy aimed to find a wide range of significant articles, whitepapers, and journals related to the topic by using well-known databases such as Google Scholar, IEEE Xplore, ResearchGate, ScienceDirect and WSU Library. A number of important search keywords, including “microservices”, “cloud computing”, “e-commerce”, and “microservices challenges”, are used to filter and improve the search result. A systematic strategy was developed to provide insights into the integration, application, and related challenges of adopting microservices in the context of cloud-based e-commerce.

– Inclusion Criteria:

- The literature was published between 2018 and 2023 to ensure the information is current and relevant.
- Focused studies on the integration, implementation, and challenges of microservices in cloud-based e-commerce.
- Only articles published in English were considered to maintain consistency and accessibility of information.

– Exclusion Criteria:

- Excluded any studies that did not specifically focus on microservices within cloud-based e-commerce environments, ensuring a targeted body of literature.
- Works that lacked empirical backing or theoretical foundation, guaranteeing the credibility and depth of the included studies.
- Duplicate studies were identified and excluded to avoid repetition and ensure a diverse range of insights.

3.3 Data Extraction

In order to assist an in-depth study, detailed attention is focused on gathering crucial data from primary sources during data extraction. An organised approach is essential making the gathered data trustworthy and comparable. The extracted data includes:

- Author(s) and year of publication
- Main objectives and methodology
- Key findings and conclusions
- Identified challenges and proposed solutions in microservices architecture.

3.4 Quality Assessment

Quality assessment is achieved through a systematic evaluation based on predefined criteria to determine the reliability and applicability of each work. The criteria include an examination of the robustness of the research methodology and establishing a foundation that verifies the integrity and validity of the findings. The reliability of the publication outlet must also be considered because it has a big impact on how well-regarded and accepted the work is in general. To make sure that the insights and conclusions reached are both immediately pertinent to and helpful to the overall discourse, each article is also examined for its degree of consistency with the review's primary issue.

3.5 Synthesis of Finding

The synthesis of findings is a crucial step, integrating both quantitative and qualitative data from a variety of sources including scholarly articles, journals, and whitepapers to offer a comprehensive understanding of microservices architecture in cloud-based e-commerce. Each selected work is individually analyzed, focusing on methodologies, conclusions, and their relevance in the current e-commerce landscape. Diverse approaches, such as in-depth case studies and broader surveys, are evaluated through a narrative synthesis method, ensuring a contextual and multifaceted analysis. This process not only addresses the primary research question but unveils patterns, contradictions, and gaps in existing literature, serving as a foundation for a richer understanding of the complex dynamics of microservices in e-commerce. The assessment integrates findings from varied methodologies and contexts, providing a comprehensive, multi-dimensional analysis that underscores prevailing trends, challenges, and opportunities in this evolving domain.

4 Findings

In the constantly evolving world of digital services, an increasing number of components combined to provide customers with complete and seamless experiences. Each component, ranging from User Experience (UX) Services to Business Logic Services, Customer Support Services, Infrastructure Services, and DevOps and CI/CD, plays a pivotal role in building, deploying, and managing effective digital applications and platforms. Each has been carefully evaluated, considering the latest technological advancements and industry trends. The highlighted features, advantages, and limitations are based on recent studies and practical experiences of professionals in the field. The goal is to provide a comprehensive insight that aids in making informed decisions for the development and management of robust, scalable, and efficient digital solutions. Each service or component is integral to ensuring that digital applications are not only user-friendly but are also secure, scalable, and easily manageable, aligning with the evolving needs of businesses and their customers (Table 1).

Table 1. Key Components in Digital Application Development.

	User Experience (UX)	Business Logic	Customer Support	Infrastructure	DevOps and CI/CD
Feature/Method	Progressive Web Apps, Cross-platform tools	AI-based recommendation, Dynamic Pricing	Push Notifications, Chatbot	Service Discovery, Microservice, Auto-Scaling	Docker, Kubernetes, CI/CD
Author/Year	[12, 13]	[14, 15]	[16, 17]	[9, 18, 19]	[11, 20]
Advantages	Fast, reliable, engaging, cost-effective, supports multiple platforms	Boosts R&D, sales, adapts to demand, maximizes profit	Enhanced engagement, increases trust	Security, scalability, fault tolerance	Lightweight virtualization, scalable
Limitations	Limited browser support, limited UX, and hardware integration issues	Integration issues, complexity in implementation	Privacy concerns, task complexity	Network control, complexity, unreliable deployment	Security concerns, setup complexity

5 Taxonomy

The proposed taxonomy not only categorises essential components of microservices architecture by also innovatively addresses gaps in existing frameworks by integrating dynamic scalability and real-time adaption capabilities required for e-commerce systems. Unlike existing models, which primarily focus on static architectural components, our taxonomy takes advantage of recent advancements in machine learning and adaptive systems to enhance e-commerce platforms’ real-time responsiveness to market changes and consumer behaviour, significantly surpassing existing models by providing a forward-looking, adaptive framework. These systems dynamically adjust offerings and operations, thereby enhancing customer engagement and satisfaction. The proposed taxonomy of e-commerce components is comprehensive and effectively reflects the complexity of the modern online business environment. Every component is designed to guarantee that businesses not only offer the best user experiences but also efficient operation, data integrity, and security in a constantly changing digital environment.

5.1 User Experience (UX)

The adoption of Progressive Web Apps (PWAs) and cross-platform tools is crucial for providing quick, reliable, and cost-effective user experiences in e-commerce. Adetunji et al. [12] describe how PWAs overcome the limits of traditional mobile development, providing consumers with a more seamless experience that does not require frequent updates or downloads. Similarly, Shah et al. [13] conduct a thorough examination of cross-platform mobile app development tools, emphasizing how these technologies allow developers to build apps that operate well across various platforms while reducing development costs and complexity. However, challenges such as limited browser support and UX integration concerns remain. This can be achieved by following best practices in

responsive design and progressive enhancement, which are crucial for developing cross-platform tools that provide a consistent experience irrespective of the device or browser used.

5.2 Business Logic

AI-driven recommendation systems and dynamic pricing are critical components in modern e-commerce platforms. Kulkov [14] investigates how artificial intelligence is transforming business operations as well as the possibilities for recommendation systems in e-commerce. Yin and Han [15] present a dynamic pricing model based on deep reinforcement learning, demonstrating how AI may improve responsiveness to changes in demand and competition, optimising pricing strategies to maximise profitability. However, integrating these systems can be complex. To solve integration complexities, microservices architecture recommended by Asrowardi et al. [9] could be utilized. This architecture enables the modular deployment of business logic components, which reduces integration concerns and allows for more efficient computation.

5.3 Customer Support

Improving customer assistance using technology like push notifications and chatbots to enhance customer engagement and trust. Sulastrı et al. [17] discuss the advantages of using push notifications into customer relationship management (CRM), emphasizing its role in increasing engagement and encouraging repeat transactions. However, privacy issues and the complexities of maintaining various customer interactions can be challenging. Cheng et al. [16] suggest that personalized chatbots, set to specific user preferences, can reduce these concerns by providing more contextually appropriate interactions.

5.4 Infrastructure

Infrastructure services are critical to the scalability and dependability of e-commerce systems. Heidari and Jafari Navimipour [18] present an in-depth study of service discovery strategies, which are essential for handling the complexity of distributed microservices. Rajput and Goyal [19] emphasise the necessity of auto-scaling solutions in ensuring system performance under variable traffic conditions. However, controlling network control and guaranteeing consistent deployments in a microservices architecture remains challenging.

5.5 DevOps and CI/CD

DevOps and continuous integration and delivery (CI/CD) are critical in the rapid development, testing, and deployment of e-commerce systems. Alanda et al. [20] examine the use of CI/CD for web-based applications, highlighting the importance of security and configuration complexity. Containerisation technologies, proposed by Bentaleb et al. [11], can help to separate and manage deployments, improving security and reducing complexity. Implementing thorough monitoring and automation technologies as part of the CI/CD pipeline may help to guarantee that deployments are consistent and secure, reducing downtime and operating risks.

6 Conclusion

The rapid development of e-commerce, facilitated by the integration of microservices and cloud applications, has marked a new era of increased scalability, flexibility, and efficiency. This study provides a considerable improvement over existing taxonomies by addressing previously overlooked complexities relating to the implementation of microservices within cloud-based e-commerce platforms. This study identifies major gaps in existing frameworks and proposes an advanced taxonomy that integrates cutting-edge technology solutions such as progressive web applications, AI-based recommendation systems, and robust CRM systems. The proposed taxonomy not only allows for dynamic and secure scaling of e-commerce platforms, but it also improves operational efficiency and customer engagement. As the e-commerce industry continues to expand, influenced by technological advances, market demands, and changing consumer preferences, this research serves as a valuable resource, directing stakeholders to make knowledgeable choices and adopt refined approaches in the constantly changing landscape of digital commerce.

References

1. Almarabeh, T., Majdalawi, Y.: Cloud computing of E-commerce. *Mod. Appl. Sci.* **13**(1), 27 (2018). <https://doi.org/10.5539/mas.v13n1p27>
2. Ramu, V.B.: Performance impact of microservices architecture. *Rev. Contemp. Sci. Acad. Stud.* **3**(6), June 2023. <https://doi.org/10.55454/rcsas.3.06.2023.010>
3. Velepucha, V., Flores, P.: A survey on microservices architecture: principles, patterns and migration challenges. *IEEE Access* **11**, 88339–88358 (2023). <https://doi.org/10.1109/ACCESS.2023.3305687>
4. Liu, G., Huang, B., Liang, Z., Qin, M., Zhou, H., Li, Z.: Microservices: architecture, container, and challenges. In: *Proceedings of the 2020 IEEE 20th International Conference on Software Quality, Reliability and Security Companion (QRS-C)*, pp. 629–635. IEEE, Macau, December 2020. <https://doi.org/10.1109/QRS-C51114.2020.00107>
5. Saboor, A., et al.: Containerized microservices orchestration and provisioning in cloud computing: a conceptual framework and future perspectives. *Appl. Sci.* **12**(12), 5793 (2022). <https://doi.org/10.3390/app12125793>
6. Abgaz, Y., et al.: Decomposition of monolith applications into microservices architectures: a systematic review. *IEEE Trans. Softw. Eng.* **49**(8), 4213–4242 (2023). <https://doi.org/10.1109/TSE.2023.3287297>
7. Kumari, A., Mohan Kumar, S.: A cloud native framework for real-time pricing in e-commerce. *Int. J. Adv. Comput. Sci. Appl.* **14**(4), 520–529 (2023). <https://doi.org/10.14569/IJACSA.2023.0140457>
8. Dobrița Ene, G.: Adaptive microservices for dynamic e-commerce: enabling personalized experiences through machine learning and real-time adaptation. *Econ. Insights Trends Chall.* **2023**(1), 95–103 (2023). <https://doi.org/10.51865/EITC.2023.01.08>
9. Asrowardi, I., Putra, S.D., Subyantoro, E.: Designing microservice architectures for scalability and reliability in e-commerce. *J. Phys. Conf. Ser.* **1450**(1), 012077 (2020). <https://doi.org/10.1088/1742-6596/1450/1/012077>
10. Yu, G., Chen, P., Zheng, Z.: *Microscaler*: Cost-effective scaling for microservice applications in the cloud with an online learning approach. *IEEE Trans. Cloud Comput.* **10**(2), 1100–1116 (2022). <https://doi.org/10.1109/TCC.2020.2985352>

11. Bentaleb, O., Belloum, A.S.Z., Sebaa, A., El-Maouhab, A.: Containerization technologies: taxonomies, applications and challenges. *J. Supercomput.* **78**(1), 1144–1181 (2022). <https://doi.org/10.1007/s11227-021-03914-1>
12. Adetunji, O., Ajaegbu, C., Otuneme, N., Omotosho, O.J.: Dawning of progressive web applications (PWA): edging out the pitfalls of traditional mobile development. **68**(1) (2020)
13. Shah, K., Sinha, H., Mishra, P.: Analysis of cross-platform mobile app development tools. In: *Proceedings of the 2019 IEEE 5th International Conference for Convergence in Technology (I2CT)*, pp. 1–7. IEEE, Bombay, March 2019. <https://doi.org/10.1109/I2CT45611.2019.9033872>
14. Kulkov, I.: The role of artificial intelligence in business transformation: a case of pharmaceutical companies. *Technol. Soc.* **66**, 101629 (2021). <https://doi.org/10.1016/j.techsoc.2021.101629>
15. Yin, C., Han, J.: Dynamic pricing model of E-commerce platforms based on deep reinforcement learning. *Comput. Model. Eng. Sci.* **127**(1), 291–307 (2021). <https://doi.org/10.32604/cmcs.2021.014347>
16. Cheng, X., Bao, Y., Zarifis, A., Gong, W., Mou, J.: Exploring consumers' response to text-based chatbots in e-commerce: the moderating role of task complexity and chatbot disclosure. *Internet Res.* **32**(2), 496–517 (2022). <https://doi.org/10.1108/INTR-08-2020-0460>
17. Sulastri, H., Sevtiyuni, P.E., Rahmatulloh, A., Sanjiwani, N.M.K.: The effect of implementing web push notification and customer relationship management in E-commerce of silver craft MSME. In: *Proceedings of the Sriwijaya International Conference on Information Technology and Its Applications (SICONIAN 2019)*. Atlantis Press, Palembang (2020). <https://doi.org/10.2991/aisr.k.200424.089>
18. Heidari, A., Jafari Navimipour, N.: Service discovery mechanisms in cloud computing: a comprehensive and systematic literature review. *Kybernetes* **51**(3), 952–981 (2022). <https://doi.org/10.1108/K-12-2020-0909>
19. Rajput, R.K.S., Goyal, D.: Auto-scaling in the cloud environment. In: Gochhait, S., Shou, D.T., Fazalbhoy, S. (eds.) *Advances in Systems Analysis, Software Engineering, and High Performance Computing*, pp. 84–98. IGI Global (2020). <https://doi.org/10.4018/978-1-7998-1294-4.ch005>
20. Alanda, A., Mooduto, H.A., Hadelina, R.: Continuous integration and continuous deployment (CI/CD) for web applications on cloud infrastructures. *JITCE J. Inf. Technol. Comput. Eng.* **6**(02), 50–55 (2022). <https://doi.org/10.25077/jitce.6.02.50-55.2022>